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Grounding device for circuit breaker

Abstract

The present disclosure relates to a grounding device for a circuit breaker, and more specifically, to a circuit breaker having a grounding device which selectively operates according to a truck. The grounding device for a circuit breaker according to an aspect of the present invention comprises: a housing installed in a distribution box or an external box; and a fixed contactor fixed to a ground truck inserted into/withdrawn from the distribution box or the external box; a moving assembly rotatably installed in the housing; a movable contactor which is coupled to and moves together with the moving assembly, and of which one end is connected to ground and the other end is connected to or separated from the fixed contactor; and a stopper assembly which is rotatably installed in the housing and restricts or allows the operation of the moving assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2021/015171, filed on Oct. 27, 2021, which claims the benefit of earlier filing date and right of priority to Korea utility model Application No. 10-2020-0153854 filed on Nov. 17, 2020, the contents of which are all hereby incorporated by reference herein in their entirety.
FIELD
(2) The present disclosure relates to a grounding device for a circuit breaker, and more particularly, to a circuit breaker having a grounding device that selectively operates according to a moving truck.
BACKGROUND
(3) In general, a circuit breaker is an electrical apparatus that is installed on a portion of a line and automatically cuts off a circuit when an overcurrent or fault current occurs in the circuit, to protect the circuit and a load.
(4) Among other circuit breakers, a vacuum circuit breaker is configured such that a vacuum interrupter (VI) disposed in the circuit breaker breaks a circuit using vacuum as extinguishing medium through an external relay, when a fault current, such as overcurrent, short-circuit, ground fault, occurs in extra-high/high-voltage distribution lines.
(5) This vacuum circuit breaker is typically installed in a switchboard in which various electrical devices are received and managed for operations or controls of power plants and substations, operations of a motor, and the like. Specifically, a circuit breaker main body is generally installed in a cradle that is fixed to the switchboard.
(6) On the other hand, for maintenance of electrical equipment, a line must first be cut off and a residual current remaining in the line must be removed. At this time, earthing switches or ground and test devices are used.
(7) The earthing switch and the grounding device are identical in that they both have a grounding function. However, the earthing switch and the grounding device are different in that the earthing switch is fixedly installed on a cradle or enclosure to perform its function, while the grounding device is a movable device separately disposed in a vacuum circuit breaker to be drawn into/out of the inside of the enclosure. In particular, the grounding device is usually used by being coupled to a truck (moving truck, moving cart), and a connection between the line and the ground must be firmly maintained during the process of drawing the grounding device into/out of the inside of the enclosure, and compatibility with the vacuum circuit breaker is also important.

(8) FIGS. **1** to **4** illustrate a grounding device according to the related art. FIG. **1** is a perspective view of a grounding device according to the related art, and FIG. **2** is a lateral view illustrating a state in which the grounding device is drawn into a switchboard.
(9) FIGS. **3** and **4** are a top perspective view and a bottom perspective view illustrating a state in which the grounding device is installed on a truck.
(10) A grounding device **1** according to the related art includes a fixing bracket **2** for supporting components and fixing them to a truck **8**, a first terminal **3** connected to a line, and a second terminal **4** connected to a grounding bus bar **7**. The grounding bus bar **7** to which the grounding device **1** is to be connected is fixedly installed in a distribution box or enclosure (not illustrated). The grounding device **1** moves between a test position and a service position together with the truck **8**, and at this time, the movement must be smooth and a contact state must also be stably maintained even during the movement. Therefore, a clip **5** is installed between the second terminal **4** and the grounding bus bar **7**. That is, the second terminal **4** is connected to the grounding bus bar **7** through the clip **5**. In addition, contact pressure is applied to the clip **5** by a contact spring **6** such that the contact (connection) is maintained in a ground state without any problems even by a repulsive force due to a fault current.
(11) On the other hand, since the grounding device **1** is used by being fixed to the truck **8**, a frame of the truck **8** needs an avoidance design for avoiding interference with the grounding bus bar **7** installed in the cradle, the distribution box, or the enclosure.
(12) An avoidance hole **9-1** is formed in a lower surface **9** of the truck **8** to prevent collision with the grounding bus bar **7** when the truck **8** is drawn into and out of the distribution box.
(13) However, the grounding device according to the related art has a structure that employs a contact method between terminals and is easy to be used at a low breaking current, but has a problem that sufficient contact pressure cannot be maintained in a system using a high breaking current. That is, the contact pressure by the clip is low.
(14) There is also a disadvantage in that such a wide avoidance hole **9-1** is required in the truck **8** since the grounding device **1** must be compatible with various types of circuit breakers in a cradle or enclosure.
(15) In addition, since the grounding device **1** occupies a considerable space in the truck **8**, there is a disadvantage in that other accessories cannot be additionally installed.
SUMMARY
(16) The present disclosure has been devised to solve those problems, and one aspect of the present disclosure is to provide a grounding device having improved contact pressure so as to be endurable even to a high breaking current.
(17) Another aspect of the present disclosure is to provide a grounding device capable of reducing a size of an avoidance hole for preventing interference among a grounding bus bar, the grounding device, and a ground truck.
(18) Still another aspect of the present disclosure is to provide a grounding device capable of reducing a space occupied thereby in a ground truck.
(19) Still another aspect of the present disclosure is to provide a grounding device capable of selectively operating a moving truck for a circuit breaker and a ground truck in an automatic manner.
(20) A grounding device for a circuit breaker in accordance with one aspect of the present disclosure may include a housing installed in a distribution box or an enclosure, a fixed contactor fixed to a ground truck that is drawn into and out of the distribution box or the enclosure, a moving assembly rotatably installed on the housing, a movable contactor coupled to the moving assembly to move together, and having one end connected to ground and another end connected to or separated from the fixed contactor, and a stopper assembly rotatably installed on the housing to restrict or allow an operation of the moving assembly.
(21) Here, the ground truck may include on a lower surface thereof a mover entry hole having a

length from a rear end to a middle portion in a front and rear direction, and an operation prevention hole formed parallel to the mover entry hole.
(22) The mover entry hole may be arranged to correspond to a position of the fixed contactor in a perpendicular direction when the ground truck enters an inside of the distribution box, and the operation prevention hole may be arranged to correspond to positions of the stopper assembly and the moving assembly in the perpendicular direction when the ground truck enters the inside of the distribution box.
(23) The ground truck may include a ground body and a moving truck that is movable while loading the ground body, and the fixed contactor may be installed on the ground body or the moving truck.
(24) The fixed contactor may include a holder fixedly installed on the ground truck, a plurality of contactors arranged in two rows on the holder with a predetermined gap therebetween, and a stator clip supporting the plurality of contactors in a surrounding manner.
(25) The grounding device may further include a fixing bracket which is fixed to the distribution box or enclosure and to which the housing is coupled.
(26) The fixing bracket may include a first operation hole through which the moving assembly is movable, and a second operation hole through which the stopper assembly is movable.
(27) The fixing bracket may further include a third operation hole through which the movable contactor is movable.
(28) The housing may include a fourth operation hole and a fifth operation hole formed through a central surface thereof to communicate with the first operation hole and the second operation hole.
(29) The grounding device may further include a support plate coupled to a side surface of the housing to support the moving assembly and the stopper assembly.
(30) The moving assembly may include a moving plate, a first shaft coupled through the moving plate, and a first lever formed on a portion of the moving plate and protruding to a top of the fixing bracket through the first operation hole.
(31) The moving assembly may further include a fixing plate which is provided on one side of the first shaft and to which the movable contactor is coupled.
(32) A first roller may be disposed on an end portion of the first lever.
(33) A landing portion with which the stopper assembly is contactable may be formed on a lower end of the moving plate.
(34) The stopper assembly may include a stopper plate, a second shaft coupled through the stopper plate, and a second lever formed on a portion of the stopper plate and protruding from a top of the fixing bracket through the second operation hole.
(35) A second roller may be disposed on an end portion of the second lever.
(36) A pressing portion which is to be brought into contact with the moving assembly may be formed on a lower end of the stopper plate.
(37) The grounding device may further include a first return spring having one end supported by a first spring fixing portion formed on the first shaft and another end supported by a second spring supporting portion formed on the stopper plate.
(38) The grounding device may further include a second return spring having one end supported by a second spring fixing portion formed on the second shaft and another end supported by a first spring supporting portion formed on the first lever.
(39) The movable contactor may be connected to the ground by a flexible connection line.
(40) A grounding device for a circuit breaker in accordance with another aspect of the present disclosure may include a stator assembly installed on a ground truck, and a mover assembly installed on a distribution box or an enclosure to be brought into contact with the stator assembly, and the mover assembly may include a housing installed on the distribution box or the enclosure, a moving assembly rotatably installed on the housing, a movable contactor coupled to the moving assembly to move together, and having one end connected to ground and another end connected to

or separated from the stator assembly, and a stopper assembly rotatably installed on the housing to restrict or permit an operation of the moving assembly.
(41) According to a grounding device for a circuit breaker according to one embodiment of the present disclosure, a connection of a ground line is made by a contact between a movable contactor and a fixed contactor, which can improve contact pressure to endure a high breaking current.
(42) A ground truck merely has an operating hole for avoiding a lever or roller of the grounding device, which can reduce a size of an avoidance hole.
(43) A main part of the grounding device is not installed on the ground truck, but is installed on a distribution box or enclosure, thereby reducing a space occupied by the ground truck.
(44) Connection and disconnection of the ground line is automatically performed according to the moving truck for the circuit breaker and the ground truck that are drawn in and out of the distribution box.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view illustrating the related art grounding device.
(2) FIG. **2** is a lateral view illustrating a state in which the grounding device of FIG. **1** is drawn into a switchboard.
(3) FIGS. **3** and **4** are a top perspective view and a bottom perspective view illustrating a state in which the grounding device of FIG. **1** is installed on a moving truck.
(4) FIG. **5** is a perspective view illustrating a state in which a grounding device for a circuit breaker according to one embodiment of the present disclosure is disposed in a switchboard.
(5) FIG. **6** is a perspective view illustrating the grounding device of FIG. **5**.
(6) FIG. **7** is an exploded perspective view of FIG. **6**.
(7) FIG. **8** is a perspective view illustrating a moving assembly and a stopper assembly of FIG. **7** viewed from different viewpoints.
(8) FIGS. **9** and **10** are a top perspective view and a bottom perspective view illustrating a moving truck to which a ground truck of FIG. **5** is applied.
(9) FIGS. **11** and **12** are views illustrating operations of the grounding device for the circuit breaker according to the one embodiment of the present disclosure, wherein FIG. **11** illustrates a state in which the moving truck for the grounding device is drawn inward, and FIG. **12** illustrates a state in which the moving truck for the circuit breaker is drawn inward.
(10) FIG. **13** is a bottom perspective view illustrating a grounding body applied to a grounding device for a circuit breaker according to another embodiment of the present disclosure.
DETAILED DESCRIPTION
(11) Hereinafter, preferred embodiments of the present disclosure will be described with reference to the accompanying drawings, so that a person skilled in the art can easily carry out the disclosure. It should be understood that the technical idea and scope of the present disclosure are not limited to those preferred embodiments.
(12) Hereinafter, a grounding device for a circuit breaker in accordance with each embodiment of the present disclosure will be described in detail with reference to the accompanying drawings. FIG. **5** is a perspective view illustrating a state in which a grounding device for a circuit breaker according to one embodiment of the present disclosure is disposed in a switchboard, FIG. **6** is a perspective view illustrating the grounding device of FIG. **5**, and FIG. **7** is an exploded perspective view of FIG. **6**. FIG. **8** is a perspective view illustrating a moving assembly and a stopper assembly of FIG. **7** viewed from different viewpoints, and FIGS. **9** and **10** are a top perspective view and a bottom perspective view illustrating a moving truck to which a ground truck of FIG. **5** is applied.
(13) A grounding device for a circuit breaker according to an embodiment of the present disclosure

includes a housing **30** installed in a portion of a distribution box **11** or a cradle, a fixed contactor **80** fixed to a ground truck **70**, **90** drawn into/out of the distribution box **11**, a moving assembly **40** rotatably installed on the housing **30**, a movable contactor **60** coupled to the moving assembly **40** and having one end connected to a ground and another end connected to or separated from the fixed contactor **80**, and a stopper assembly **50** rotatably installed on the housing **30** to restrict or allow an operation of the moving assembly **40**.

(14) In the grounding device for the circuit breaker according to the one embodiment of the present disclosure, a stator assembly part and a mover assembly part are separately installed. The stator assembly part is installed on the ground truck **70**, **90**, and the mover assembly part is installed on the distribution box **11** or an enclosure (here, the enclosure refers to a cradle, a grounding box, or a grounding chamber). The stator assembly part may simply include the fixed contactor **80**.

(15) First, a ground and test truck (hereinafter, referred to as a ground truck) **73**, **90** will be described. FIGS. **9** to **10** illustrate the ground truck.

(16) The ground truck **70**, **90** may include a ground body **90** and a moving truck **70**.

(17) Referring to FIG. **13**, the ground body **90** may be configured similarly to a body of a vacuum circuit breaker. At this time, a breaker such as a vacuum interrupter is not provided inside the ground body. The ground body **90** functions as a current connection passage. The ground body **90** may include a body part **91**, a terminal part **93** disposed at the rear of the body part **91**, and a bus bar part **95** disposed below the body part **91**. The bus bar part **95** is connected to the terminal part **93** and is connected to a main circuit when the ground truck **70**, **90** drawn into the distribution box **11** or the enclosure. Depending on an embodiment, the fixed contactor **80** may be installed on a portion of the bus bar part **95**.

(18) The moving truck **70** may be formed similarly to a truck of a vacuum circuit breaker. The moving truck **90** may carry the ground body **90**.

(19) A lower surface **76** of the moving truck **70** is provided with a mover entry hole **72** formed in a length from a rear end to a middle portion, and an operation prevention hole **74** formed parallel to the mover entry hole **72**.

(20) The mover entry hole **72** is formed in a shape that the lower surface **76** of the moving truck **70** is partially cut. The mover entry hole **72** is formed in a length from a rear portion to a middle portion of the moving truck **70**. The mover entry hole **72** is formed in a movement direction of the moving truck **70**, that is, in a front-rear direction. The mover entry hole **72** is disposed to correspond to a position of the fixed contactor **80** when the moving truck **70** enters the inside of the distribution box **11**. When the moving truck **70** enters the distribution box **11** or the cradle, the movable contactor **60** is inserted through the mover entry hole **72**. The movable contactor **60** is inserted into the fixed contactor **80** to be in contact with the fixed contactor **80**.

(21) The operation prevention hole **74** is formed in a shape that the lower surface of the moving truck **70** is partially cut. The operation prevention hole **74** is formed in a length from a rear portion to a middle portion of the moving truck **70**. The operation prevention hole **74** is formed next to the mover entry hole **72** to be in parallel to the mover entry hole **72**. The operation prevention hole **74** is formed in a movement direction of the moving truck **70**, that is, in the front-rear direction. The operation prevention hole **74** is placed at positions of the stopper assembly **50** and the moving assembly **40** in a perpendicular direction when the moving truck **70** enters the inside of the distribution box **11**. That is, the stopper assembly **50** and the moving assembly **40** are inserted into the operation prevention hole **74** and remain stationary without movement. Therefore, the movable contactor **60** is in a fixed state and is inserted into the fixed contactor **80**.

(22) The stator assembly part, that is, the fixed contactor **80** is installed on the moving truck **70**. That is, the fixed contactor **80** is installed on the ground body **90** or the moving truck **70**.

(23) For example, the fixed contactor **80** is installed on an upper surface **78** of the moving truck **70** (see FIG. **9**). The fixed contactor **80** is installed on the moving truck **70** to be brought into contact with the movable contactor **60**. Accordingly, the ground is connected and a current remaining in a

circuit of the distribution box **11** is discharged. The fixed contactor **80** is connected to the line.
(24) As another example, the fixed contactor **80** is installed on the ground body **90**. Referring to FIG. **13**, the fixed contactor **80** may be installed on the bus bar part **95** beneath the ground body **90**.
(25) The fixed contactor **80** may be formed in various shapes according to embodiments. As an example, referring to FIG. **5**, the fixed contactor **80** applied to this embodiment may include a holder **81** formed in a shape like "T" fixedly installed on the upper surface **78** of the moving truck **70**, a plurality of contactors **83** arranged in two rows with a gap therebetween on the holder **81**, and a stator clip **85** surrounding and supporting the plurality of contactors **83**. The plurality of contactors **83** arranged in the two rows face each other with a predetermined gap, so that the movable contactor **60** is coupled by engagement between the plurality of contactors **83**. That is, the movable contactor **60** receives contact pressure by the plurality of contactors **83**. That is, since the grounding connection of the grounding device according to the one embodiment of the present disclosure is made by a contact between the fixed contactor **80** and the movable contactor **60**, contact pressure can be provided sufficiently. Therefore, it may be applicable even to a high voltage device as well as a low voltage device.
(26) Hereinafter, the mover assembly part installed on the distribution box **11**, the enclosure, or the cradle will be described.
(27) The mover assembly **20** is installed on a bottom surface of the distribution box **11** (a surface on which the moving truck or the ground truck moves while being in contact).
(28) The mover assembly **20** includes a fixing bracket **21** disposed on a portion of the distribution box **11**, a housing **30** fixed to the fixing bracket **21**, a moving assembly **40** rotatably installed on the housing **30**, a movable contactor **60** coupled to the moving assembly **40** and having one end connected to the ground, and a stopper assembly **50** restricting or allowing the operation of the moving assembly **40**.
(29) The fixing bracket **21** has an accommodating portion **22** to accommodate members of the mover assembly **20**. For example, the fixing bracket **21** may be formed in a '⊏' shape.
(30) The fixing bracket **21** includes a first operation hole **23** through which the moving assembly **40** is movable, and a second operation hole **24** through which the stopper assembly **50** is movable. The moving assembly **40** is partially exposed through the first operation hole **23** and the stopper assembly **50** is partially exposed through the second operation hole **24**. That is, in a normal state, the stopper assembly **50** and the moving assembly **40** protrude above an upper surface of the fixing bracket **21**. The first operation hole **23** and the second operation hole **24** may be connected into a single hole.
(31) The fixing bracket **21** includes a third operation hole **25** through which the movable contactor **60** is movable.
(32) A housing **30** is provided. The housing **30** is provided to install the stopper assembly **50** and the moving assembly **40** thereon. The housing **21** may be formed in a '⊏' shape. The housing **30** is seated in the accommodating portion **22** of the fixing bracket **21**.
(33) A fourth operation hole **31** and a fifth operation hole **32** are formed through a central surface of the housing **30** to communicate with the first operation hole **23** and the second operation hole **24**.
(34) The housing **30** includes a first shaft hole **34** and a second shaft hole **35** formed through one side surface thereof to support a first rotational shaft **41** of the moving assembly **40** and a second rotational shaft **51** of the stopper assembly **50**, respectively.
(35) The housing **30** also includes a first through hole **36** and a second through hole **37** formed through another side surface thereof, such that the first rotational shaft **41** of the moving assembly **40** and the second rotational shaft **51** of the stopper assembly **50** are inserted therethrough, respectively.
(36) A support plate **26** coupled to the another side surface of the housing **30** is provided. The support plate **26** includes a third shaft hole **27** and a fourth shaft hole **28** formed therethrough to support the first rotational shaft **41** of the moving assembly **40** and the second rotational shaft **51** of

the stopper assembly **50**.
(37) A moving assembly **40** is provided. The moving assembly **40** is provided to selectively move the movable contactor **60** according to a type of moving truck. Detailed drawings of the moving assembly **40** may refer to FIGS. **7** and **8**.
(38) The moving assembly **40** is provided with a first shaft **41** acting as a rotational shaft. Both ends of the first shaft **41** are supported by the first shaft hole **34** and the third shaft hole **27**, respectively.
(39) The first shaft **43** is coupled to the moving plate **43**. The moving plate **43** may be formed in a 'custom character' shape. A first lever **44** is provided on one end (upper end) of the moving plate **43**.
(40) The first lever **44** protrudes to the outside (top) of the fixing bracket **21** through the first operation hole **23**. When the first lever **44** is pressed, the moving plate **43** rotates centering on the first shaft **41**, and the movable contactor **60** coupled to the moving assembly **40** also rotates.
(41) A first roller **47** may be disposed on an end portion of the first lever **44**. An impact applied to the first lever **44** is alleviated by the roller **47**. The roller **47** is rolled in contact with a rear surface **171** and a lower surface **176** of the moving truck **170**, which can make the moving assembly **40** moved smoothly and improve durability of the moving assembly **40**.
(42) A first spring supporting portion **48** supporting a second return spring **66** is provided on a lower portion of the first lever **44**.
(43) A landing portion **45** with which the stopper assembly **50** can come into contact is formed at another end (lower end) of the moving plate **43**. The landing portion **45** may be formed flat.
(44) The first shaft **41** is provided with a fixing plate **49** capable of fixing the movable contactor **60**. The movable contactor **60** is fixed to the fixing plate **49** by screwing or the like.
(45) A first spring fixing portion **46** supporting the first return spring **65** is provided on another end of the first shaft **41**.
(46) A stopper assembly **50** is provided. The stopper assembly **50** is provided to restrict or allow the movement of the moving assembly **40**.
(47) A second shaft **51** acting as a rotational shaft is disposed in the stopper assembly **50**. Both ends of the second shaft **51** are supported by the second shaft hole **35** and the fourth shaft hole **28**, respectively.
(48) A stopper plate **53** is coupled to the second shaft **51**. The stopper plate **53** may be a flat plate in a shape like 'V' forming an obtuse angle. A second lever **54** is provided on one end (upper end) of the stopper plate **53**.
(49) The second lever **54** protrudes to the outside (top) of the fixing bracket **21** through the second operation hole **24**. When the second lever **54** is pressed, the stopper plate **53** rotates centering on the second shaft **51**.
(50) A second roller **57** may be disposed on an end portion of the second lever **54**. An impact applied to the second lever **54** is alleviated by the second roller **57**. The second roller **57** is rolled in contact with the rear surface **171** and the lower surface **176** of the moving truck **170**, which can make the stopper assembly **50** moved smoothly and improve durability of the stopper assembly **40**.
(51) A second spring fixing portion **58** supporting a second return spring **66** is provided on the second shaft **51**.
(52) A pressing portion **55** that can be brought into contact with the landing portion **45** is formed on another end (lower end) of the stopper plate **53**. When the pressing portion **55** is brought into contact with the landing portion **45**, the movement of the moving assembly **40** is restricted so that the movable contactor **60** remains in a fixed state.
(53) A second spring supporting portion **56** supporting the first return spring **65** is provided at another end (lower end) of the stopper plate **53**.
(54) One end of the first return spring **65** is supported by the first spring fixing portion **46** of the first shaft **41** and another end is supported by the second spring supporting portion **56** of the

stopper plate **53**.
(55) One end of the second return spring **66** is supported by the second spring fixing portion **58** of the second shaft **51** and another end is supported by the first spring supporting portion **48** of the first lever **44**.
(56) The movable contactor **60** is made of a material with good electrical conductivity. The movable contactor **60** is fixedly coupled to the support plate **49** and moves along with the rotation of the moving assembly **40**.
(57) The movable contactor **60** is connected to the ground by a flexible connection line **73** such as a braided wire.
(58) Hereinafter, operations of the grounding device for the circuit breaker according to the one embodiment of the present disclosure will be described.
(59) FIG. **11** illustrates a state in which the moving truck **70**, on which the grounding device according to the one embodiment is installed, is drawn into the distribution box **11**.
(60) Since the moving truck **70** is provided with the mover entry hole **72**, it does not interfere with the movable contactor **60** during its movement. In addition, since the moving truck **70** is provided with the operation prevention hole **74**, it does not interfere with the movable contactor **60** during its movement. Therefore, the fixed contactor **80** attached to the moving truck **70** is fitted into the movable contactor **60** installed on the distribution box **11**, such that a grounding circuit is connected and residual current on the circuit is removed accordingly.
(61) FIG. **12** illustrates a state in which a general moving truck **170** for moving a circuit breaker (hereinafter, referred to as a moving truck for a circuit breaker) is drawn into the distribution box **11**. That is, it shows the operation when the circuit breaker (vacuum breaker) is drawn in.
(62) The moving truck **170** for the circuit breaker does not have the mover entry hole **72** or the operation prevention hole **74**. Therefore, when the moving truck **170** for the circuit breaker enters, the rear surface **171** of the moving truck **170** for the circuit breaker pushes the second lever **54** or the second roller **57** of the stopper assembly **50** so that the stopper assembly **50** is rotated. Since the pressing portion **55** of the stopper assembly **50** is out of a motion range of the moving assembly **40**, the moving assembly **40** can move freely. As the moving truck **170** for the circuit breaker further enters, the first lever **44** or the first roller **47** of the moving assembly **40** is pushed such that the moving assembly **40** is rotated. As the moving assembly **40** is rotated downward, the movable contactor coupled to the moving assembly **40** is also rotated downward so as not to interfere with the moving truck **170** for the circuit breaker. A ground line is not connected when the general moving truck **170** enters.
(63) In a grounding device for a circuit breaker according to one embodiment of the present disclosure, a connection of a ground line is made by a contact between a movable contactor and a fixed contactor, which can improve contact pressure to endure a high breaking current.
(64) A ground truck merely has an operation hole for avoiding a lever or roller of the grounding device, which can reduce a size of an avoidance hole.
(65) A main part of the grounding device is not installed on the ground truck, but is installed on a distribution box or enclosure, thereby reducing a space occupied by the ground truck.
(66) Connection and disconnection of the ground line is automatically performed according to the moving truck for the circuit breaker and the ground truck that are drawn in and out of the distribution box.
(67) While the invention has been shown and described with reference to the foregoing preferred embodiments thereof, it will be understood by those skilled in the art that various changes and modifications may be made without departing from the scope of the invention as defined by the appended claims. Therefore, the embodiments disclosed in the present disclosure are not intended to limit the scope of the present disclosure but are merely illustrative, and it should be understood that the scope of the technical idea of the present disclosure is not limited by those embodiments. That is, the scope of protection of the present disclosure should be construed according to the

appended claims, and all technical ideas within the scope of equivalents thereof should be construed as being included in the scope of the present disclosure.

Claims

1. A grounding device for a circuit breaker, the device comprising: a housing installed in a distribution box or an enclosure; a fixed contactor fixed to a ground truck that is drawn into and out of the distribution box or the enclosure; a moving assembly rotatably installed on the housing; a movable contactor coupled to the moving assembly to move together, and having one end connected to ground and another end connected to or separated from the fixed contactor; and a stopper assembly rotatably installed on the housing to restrict or allow an operation of the moving assembly, wherein the ground truck includes on a lower surface thereof a mover entry hole having a length from a rear end to a middle portion in a front and rear direction, and an operation prevention hole formed parallel to the mover entry hole.
2. The grounding device of claim 1, wherein the mover entry hole is arranged to correspond to a position of the fixed contactor in a perpendicular direction when the ground truck enters an inside of the distribution box, and wherein the operation prevention hole is arranged to correspond to positions of the stopper assembly and the moving assembly in the perpendicular direction when the ground truck enters the inside of the distribution box.
3. The grounding device of claim 1, wherein the ground truck includes a ground body and a moving truck that is movable while loading the ground body, and wherein the fixed contactor is installed on the ground body or the moving truck.
4. The grounding device of claim 3, wherein the fixed contactor comprises: a holder fixedly installed on the ground truck; a plurality of contactors arranged in two rows on the holder with a predetermined gap therebetween; and a stator clip supporting the plurality of contactors in a surrounding manner.
5. The grounding device of claim 1, further comprising a fixing bracket which is fixed to the distribution box or enclosure and to which the housing is coupled.
6. The grounding device of claim 5, wherein the fixing bracket includes a first operation hole through which the moving assembly is movable, and a second operation hole through which the stopper assembly is movable.
7. The grounding device of claim 6, wherein the fixing bracket further includes a third operation hole through which the movable contactor is movable.
8. The grounding device of claim 6, wherein the housing includes a fourth operation hole and a fifth operation hole formed through a central surface thereof to communicate with the first operation hole and the second operation hole.
9. The grounding device of claim 6, wherein the moving assembly comprises: a moving plate; a first shaft coupled through the moving plate; and a first lever formed on a portion of the moving plate and protruding to a top of the fixing bracket through the first operation hole.
10. The grounding device of claim 9, wherein the moving assembly further comprises a fixing plate which is provided on one side of the first shaft and to which the movable contactor is coupled.
11. The grounding device of claim 9, wherein a first roller is disposed on an end portion of the first lever.
12. The grounding device of claim 9, wherein a landing portion with which the stopper assembly is contactable is formed on a lower end of the moving plate.
13. The grounding device of claim 9, wherein the stopper assembly comprises: a stopper plate; a second shaft coupled through the stopper plate; and a second lever formed on a portion of the stopper plate and protruding from a top of the fixing bracket through the second operation hole.
14. The grounding device of claim 13, wherein a second roller is disposed on an end portion of the second lever.

15. The grounding device of claim 13, wherein a pressing portion which is to be brought into contact with the moving assembly is formed on a lower end of the stopper plate.
16. The grounding device of claim 13, further comprising a first return spring having one end supported by a first spring fixing portion formed on the first shaft and another end supported by a second spring supporting portion formed on the stopper plate.
17. The grounding device of claim 16, further comprising a second return spring having one end supported by a second spring fixing portion formed on the second shaft and another end supported by a first spring supporting portion formed on the first lever.
18. The grounding device of claim 1, further comprising a support plate coupled to a side surface of the housing to support the moving assembly and the stopper assembly.
19. The grounding device of claim 1, wherein the movable contactor is connected to the ground by a flexible connection line.
20. A grounding device for a circuit breaker, the device comprising: a stator assembly installed on a ground truck; and a mover assembly installed on a distribution box or an enclosure to be brought into contact with the stator assembly, wherein the mover assembly comprises: a housing installed on the distribution box or the enclosure; a moving assembly rotatably installed on the housing; a movable contactor coupled to the moving assembly to move together, and having one end connected to ground and another end connected to or separated from the stator assembly; and a stopper assembly rotatably installed on the housing to restrict or permit an operation of the moving assembly, wherein the ground truck includes on a lower surface thereof a mover entry hole having a length from a rear end to a middle portion in a front and rear direction, and an operation prevention hole formed parallel to the mover entry hole.